SPECIFICATION

Cross-Reference to Related Applications

This application is related to and incorporates by reference the entirety of U.S. Patent Application No. 19/746,749, titled "A MONOLITHIC BIDIRECTIONAL ACTIVE-MATRIX DISPLAY WITH ENDOGENOUS SELF-FEEDBACK IMAGING (P.E.R.W.I.N.)," filed on July 20, 2026 (Confirmation #5096, Patent Center #78998550).

Background of the Invention

Covert surveillance devices are frequently disguised within common household objects, such as smoke detectors, clock radios, and light bulbs. However, conventional covert light bulb cameras rely on traditional, discrete CMOS camera modules hidden behind a small physical aperture or a one-way mirrored section of the bulb's casing. These discrete camera lenses are highly vulnerable to detection by counter-surveillance sweeps using optical lens finders, thermal imaging (due to localized heat from the camera SoC), or simple physical inspection of the bulb's surface. Furthermore, the discrete camera occupies internal volume, reducing the available space for illumination hardware and leading to asymmetrical or suspicious light dispersion. There is a critical need for a covert imaging device that is physically and optically integrated into the light-emitting surface itself, rendering the camera absolutely undetectable by conventional counter-surveillance techniques.

Summary of the Invention

The present invention resolves the vulnerabilities of discrete covert cameras by applying a bidirectional active-matrix architecture (P.E.R.W.I.N.) to a three-dimensional, omnidirectional illumination form factor.

Rather than hiding a camera inside a bulb, the invention transforms the entire light-emitting surface of the bulb into the camera. The apparatus utilizes a flexible polymer substrate (such as polyimide) that is physically shaped or thermoformed to map to the interior surface of a standard bulb enclosure (e.g., an A19 bulb, BR30 flush mount, or GU10 floodlight).

Deposited onto this curved substrate is an interleaved matrix of self-emissive subpixels (capable of outputting single-color white light, or variable RGB light) and Organic Photodetector (OPD) sensor subpixels.

The device is driven by a logic controller housed in the base of the bulb (e.g., within the E26 screw base). This controller executes a high-frequency, de-phased duty cycle. For example, operating at 120 Hz, the controller globally deactivates the emissive subpixels for a 500μs interval. During this precise dark window, the OPD sensors are activated to integrate the ambient light reflecting off the subjects in the room. Because the integration window is extremely short and the switching frequency is well above the human flicker fusion threshold, the bulb appears to emit a continuous, high-lumen output identical to a standard commercial LED bulb.

The curved nature of the substrate provides an inherent 360° or hemispherical field of view. The raw sensor data is processed by a computational deconvolution matrix to reconstruct a panoramic outward image, which is subsequently exfiltrated via integrated wireless communication (e.g., Wi-Fi, Bluetooth) or modulated directly over the AC power mains using Power Line Communication (PLC).